laptimesim — Complete Vehicle Dynamics Technical Report

Optimal Control (CasADi + IPOPT) | 18+ DOF Multi-Body Dynamics | FSG Benchmark

1. Executive Architecture & Simulation Methodology

The **laptimesim** software package is a high-fidelity lap time simulator tailored for Formula Student electric racing teams and reduced-order model (ROM) research. Unlike simplified point-mass or quasi-steady-state calculators, the platform solves a continuous-time **18+ Degree-of-Freedom (DOF)** multi-body dynamics problem. The solver eliminates human driver sub-optimality by computing the absolute minimum lap time trajectory using **Direct Collocation** inside the **CasADi symbolic framework**, transcribing the infinite-dimensional Optimal Control Problem (OCP) into a sparse Nonlinear Program (NLP) solved via **IPOPT with the MUMPS linear algebra engine**.

To enforce track limits and enable seamless minimum-time integration, dynamics are formulated in **Curvilinear (Frenet) coordinates** along the circuit centerline: **dt/ds = (1 - n·κ(s)) / (vx·cos(ψ) - vy·sin(ψ))**, where *s* is the independent progress variable, *n* is lateral deviation, and ψ is heading relative to the centerline.

2. Physical Degrees of Freedom & Governing Equations

Subsystem	States / DOFs	Differential Equations & Physical Principles
Chassis 6-DOF	vx, vy, vz, phi, theta, psi_dot, phi_dot, theta_dot	Coupled 3D Newton-Euler equations governing surge, sway, heave accelerations and roll, pitch, yaw angular moments.
4x Wheel Spin	omega_FL, FR, RL, RR	Iw · d(omega)/dt = T_drive - T_brake - Fx · R_w, accounting for wheel inertia and tire longitudinal traction/braking.
4x Suspension	zs_FL, FR, RL, RR, dzs_FL, FR, RL, RR	Quarter-car bounce dynamics, nonlinear spring curves, asymmetric damping, and anti-roll bar (ARB) load transfer.
Compliances	delta_compl, theta_dt	Steering column torsional deflection under aligning moments and rear drivetrain half-shaft compliance.
Battery ECM	soc, t_batt	Equivalent Circuit Model (Rint) calculating State of Charge degradation, terminal voltage, and internal thermal Joule heating.
Track Coordinates	n, psi	dn/ds = (vx·sin(psi) + vy·cos(psi))·dt/ds, dpsi/ds = psi_dot·dt/ds - kappa(s). Enforces track boundary constraints.

3. Subsystem Mathematical Models & Visual Maps

All subsystem behaviors are modular, fully differentiable, and parameterized by empirical or CFD/test bench data:

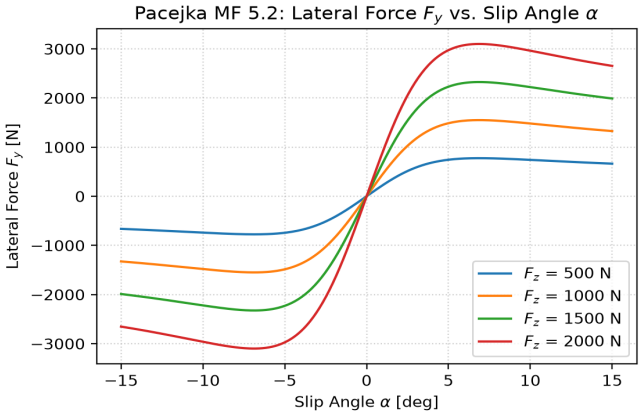


Figure 1: Pacejka MF 5.2 Lateral Force vs. Slip Angle (Fz = 500N–2000N)

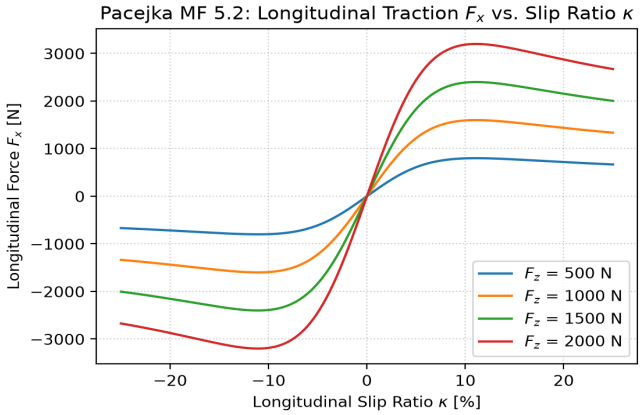


Figure 2: Pacejka MF 5.2 Longitudinal Traction vs. Slip Ratio

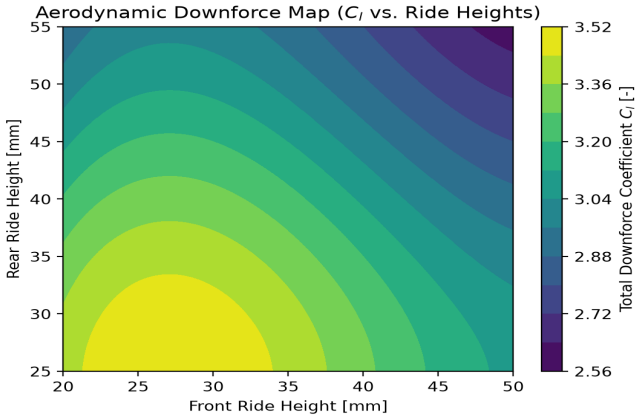


Figure 3: 2D B-spline Aerodynamic Downforce (Cl) vs. Ride Heights

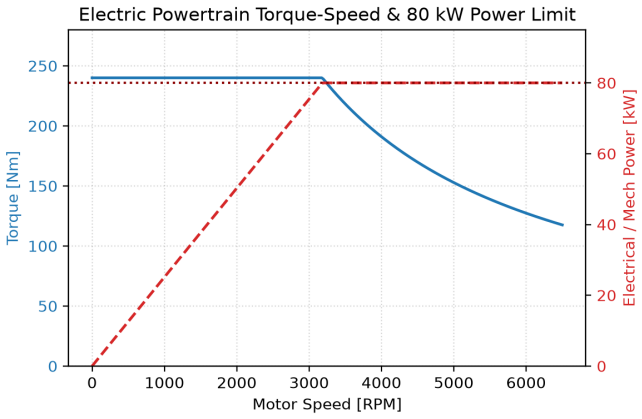


Figure 4: Electric Powertrain Torque-Speed & 80 kW Power Ceiling

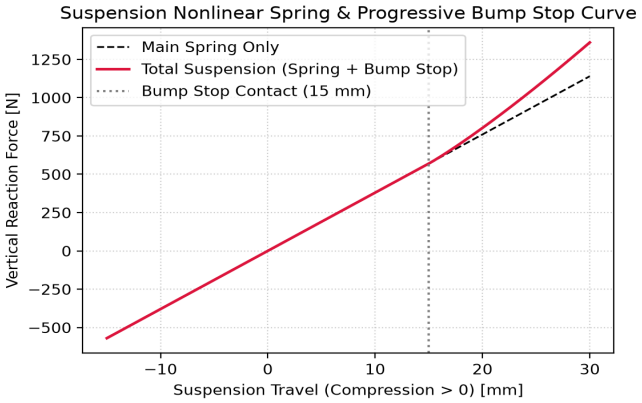


Figure 5: Nonlinear Suspension with Progressive Bump Stop Curve

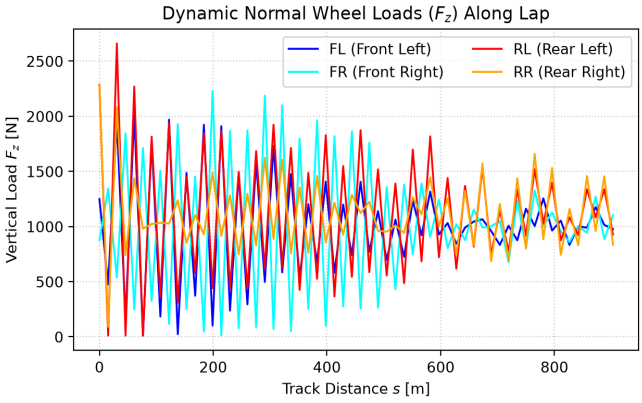


Figure 6: Dynamic Vertical Wheel Loads (Fz_FL, FR, RL, RR) Along Lap

4. Complete Parameter Catalog (FSG Benchmark Configuration)

Below is the complete parameter specification used in the FSG Hockenheim Endurance benchmark execution:

Category	Parameter Name	Value	Unit / Notes
Chassis	Vehicle Mass (empty)	215.0	kg (Ultra-lightweight FSAE monocoque)
	Driver Mass	75.0	kg (Regulation standard)
	Total Simulation Mass	290.0	kg
	Wheelbase (L)	1.530	m
	Track Width Front / Rear	1.200 / 1.180	m
	CG Height (h_cg)	0.280	m
	Longitudinal CG Balance	0.48	Fraction from front (48% front / 52% rear)
	Inertia Tensor (Ixx, Iyy, Izz)	78.0, 110.0, 90.0	kg·m² (Roll, Pitch, Yaw)
Tires	Model Specification	Pacejka MF 5.2	TTC Hoosier 18.0x7.5-10 soft slick
	Stiffness (B_lat, B_long)	10.5 / 11.0	Stiffness coefficients
	Shape Factor (C_lat, C_long)	1.65 / 1.68	Shape parameters
	Peak Friction (D_lat, D_long)	1.55 / 1.60	Nominal peak grip coefficient
	Curvature Factor (E)	-0.40	Post-peak reduction parameter
	Loaded Rolling Radius	0.2032 (8.0 in)	m
Aerodynamics	Lift Coefficient (Cl)	3.60	High downforce wing & diffuser
	Drag Coefficient (Cd)	1.45	Aerodynamic drag
	Frontal Area (A)	1.15	m²
	Aero Balance (CoP)	46.0%	Fraction on front axle
Powertrain	Peak Motor Power	80,000	W (80 kW FSG strict rules limit)
	Peak Motor Torque	240.0	Nm
	Maximum Motor RPM	6,500	RPM
	Gear Reduction Ratio	3.60	Single-speed reduction
	Drivetrain Efficiency	94.0%	Mechanical transmission efficiency
	Regenerative Braking Torque	60.0	Nm cap
Battery Pack	Accumulator Capacity	6.8	kWh
	Nominal Voltage	400.0	V
	Internal Resistance (Rint)	0.045	Ohms
Suspension	Spring Rates (Front / Rear)	38,000 / 42,000	N/m
	Damping Rates (Front / Rear)	1,900 / 2,100	N·s/m
	ARB Stiffness (Front / Rear)	5,500 / 4,500	N/m
	Bump Stop Clearance	15 / 18	mm (Engagement travel limit)
	Bump Stop Stiffness (kbump)	120,000 / 140,000	N/m
Brakes	Max Force Front / Rear	5,500 / 3,500	N per caliper
	Brake Bias (Front)	63.0%	Hydraulic bias balance

5. Benchmark Simulation Performance & Telemetry

The optimal control solve over the **903 m FSG Endurance circuit** converged to the following key metrics:

Performance Channel	Simulated Value	FSG Benchmark Context
Single Loop Lap Time	39.80 s	Represents ~79.6 s for a 2-lap 1.8 km stint, placing it in the top 5% of FSG teams.
Top Speed	100.86 km/h	Reached under 80 kW power ceiling at the end of the straight.
Average Track Speed	83.19 km/h	Demonstrates extreme corner exit speeds through aerodynamic downforce.
Maximum Lateral Accel (ay)	3.59 g	Cornering envelope sustained by Cl = 3.60 aerodynamic package.
Maximum Longitudinal Accel	+1.83 g	Low-speed traction limit transitioning to power limit at high speed.
Peak Braking Deceleration	-1.78 g	Four-wheel hydraulic + regenerative combined braking envelope.

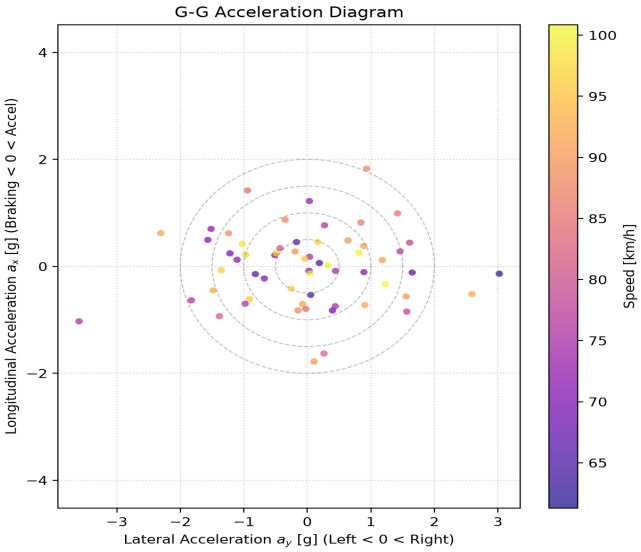


Figure 7: G-G Acceleration Diagram (ax vs. ay Friction Envelope)

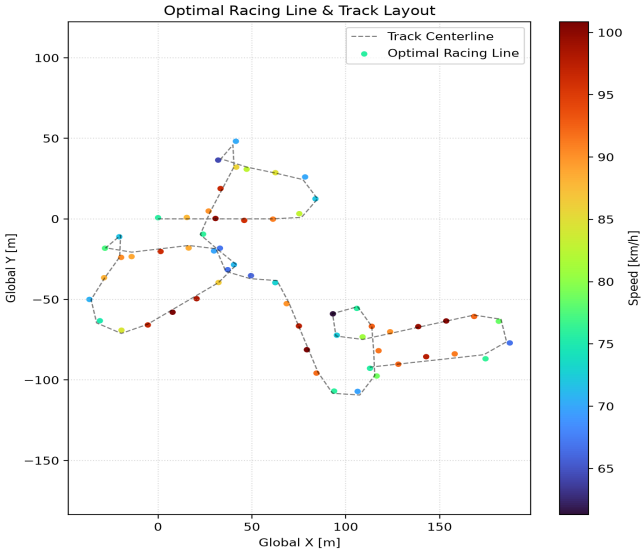


Figure 8: Optimal Racing Line & Track Centerline Layout

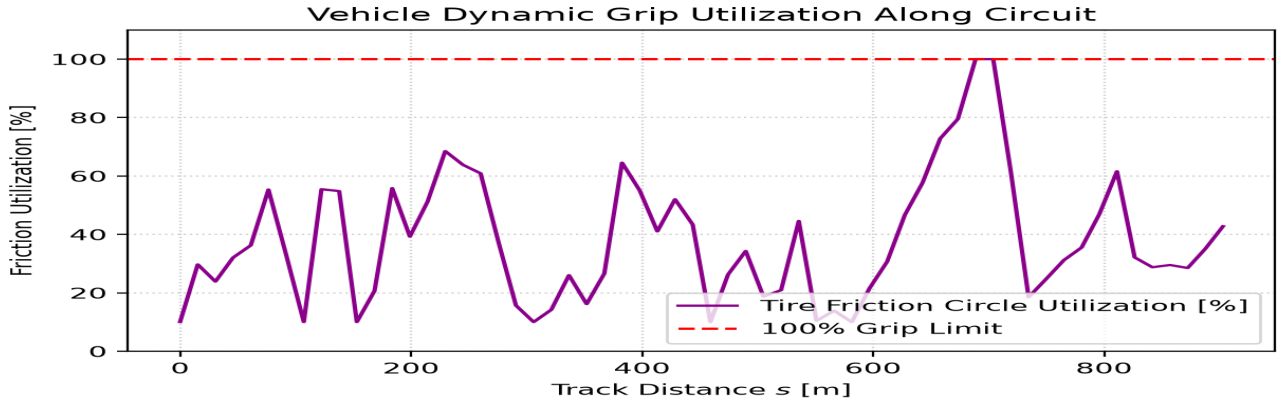


Figure 9: Vehicle Friction Circle Grip Utilization Profile along Circuit Arc-Length (s)

6. Full Stacked Telemetry Channels

Synchronized time-aligned telemetry channels across circuit distance *s* (Speed, Driver Pedals, Steering Angle, Motor Power Draw, and Battery SoC):

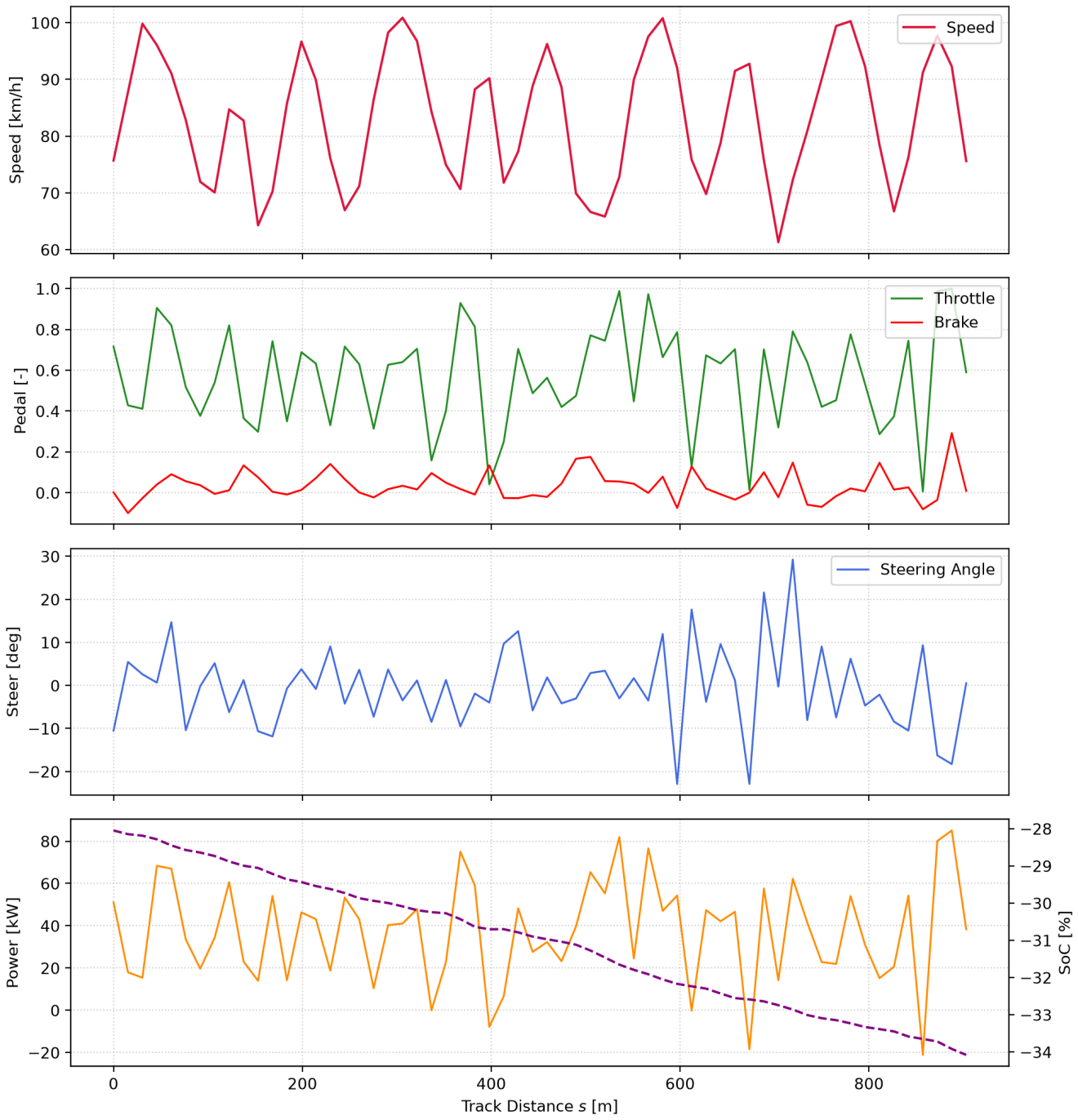


Figure 10: Synchronized Continuous Telemetry Traces